

# **Documentation on Agents and Their Types**

## **1. What is an Agent?**

In simple terms, an agent is something that can act independently based on its perception of the environment. An agent takes input from its surroundings (called percepts) and then performs some action.

In computer science, particularly in the field of Artificial Intelligence (AI), an agent is a program or system that can make decisions autonomously to achieve a specific goal.

### **Examples:**

- A robot that moves towards light is an agent.
- A chatbot that answers questions is also an agent.
- Even a simple thermostat that controls temperature can be considered an agent.

-So basically, an agent can sense, think, and act.

## **2. Structure of an Agent**

Every agent generally has two main parts:

**1. Sensors** – These help the agent observe or collect information from its environment.

**2. Actuators** – The agent uses these to perform actions or respond based on what it has sensed.

### **Example:**

A self-driving car uses cameras and sensors to see the road and its surroundings, and utilizes steering to act and move according to the road conditions.

### 3. Types of Agents

There are mainly five types of agents in Artificial Intelligence:

#### a) Simple Reflex Agent

- Works only based on the current input.
- Follows simple if-condition-then-action rules.
- Does not think about past actions or future outcomes.

**Example:** A vacuum cleaner that turns when it hits an obstacle.

#### b) Model-Based Reflex Agent

- Holds some internal information (a model) about the world.
- Uses this model to make informed decisions based on both current and past information.

**Example:** A robot that remembers where it found obstacles earlier and avoids them next time.

#### c) Goal-Based Agent

- Acts to achieve a specific goal or target.
- Chooses its actions based on what will bring it closer to the goal.

**Example:** A GPS navigation system finds the shortest route to reach a destination.

#### d) Utility-Based Agent

- Focuses on finding not just any solution, but the best or most efficient one.
- Uses a utility function to measure how good or bad a particular action or state is.

**Example:** A delivery robot that chooses the fastest and safest route instead of just any available path.

#### **e) Learning Agent**

- It can learn from past experiences and improve its performance over time.
- Adjusts its actions based on feedback or results from previous tasks.

**Example:** An AI in games that becomes smarter after playing multiple rounds.